The Islamia University of Bahawalpur

TERM PROPOSAL | Research Methods

Explainable Identification of Human vs. Large Language Model (LLM) Generated Text: A Review

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EXECUTIVE SUMMARY

The rapid advancement of Large Language Models (LLMs) has significantly transformed the landscape of digital content creation, enabling the generation of highly coherent and human-like text. While these developments offer substantial benefits, they also introduce serious concerns related to academic integrity, misinformation, and the authenticity of online content. As a result, distinguishing between human-written and AI-generated text has become an important research challenge in natural language processing.

Although existing detection approaches—ranging from traditional machine learning techniques to advanced transformer-based models—have demonstrated promising performance, most of these systems operate as “black boxes.” This lack of transparency limits users’ ability to understand and trust the decisions made by such models, especially in sensitive domains like education, journalism, and research evaluation.

This proposal presents a structured plan for a review article that explores the intersection of human vs. LLM-generated text detection and Explainable Artificial Intelligence (XAI). The primary objective is to critically analyze existing detection methods while examining how explainability techniques, such as LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations), contribute to making these systems more transparent and interpretable.

Specifically, the research aims to (i) categorize existing detection paradigms, including traditional, deep learning, and stylometric approaches, (ii) evaluate the effectiveness of XAI methods in identifying key linguistic and stylistic features, (iii) assess the generalization capability of detection models across different LLMs and domains, and (iv) identify existing research gaps and limitations in current approaches.

By synthesizing recent literature **(2023–2026**), this study seeks to provide a comprehensive and coherent understanding of explainable AI-based text detection. The findings of this research are expected to contribute toward the development of more reliable, transparent, and trustworthy detection systems, while also offering valuable insights for researchers, educators, and practitioners dealing with AI-generated content.

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1. INTRODUCTION

The evolution of generative AI, particularly Large Language Models (LLMs) such as GPT-4, Llama, and Claude, has revolutionized content creation while simultaneously introducing complex challenges in distinguishing between human-authored and machine-generated text. As these models grow increasingly sophisticated, their outputs have become remarkably fluent and contextually coherent, rendering manual identification unreliable and necessitating the development of robust, automated detection tools. This challenge is especially acute in trust-sensitive domains such as academia, journalism, and legal proceedings, where the provenance of text carries significant ethical and institutional weight.

However, accurate detection alone is no longer sufficient. There is a growing demand for explainability — the ability to understand and justify why a piece of text is classified as AI-generated. Without such transparency, detection systems risk being dismissed as unreliable, particularly when their outputs carry consequential implications for individuals or institutions. Among the available interpretability frameworks, post-hoc local explanation methods — specifically LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations) — have emerged as the most widely adopted tools in text classification research. Their model-agnostic nature makes them applicable across both traditional machine learning and deep learning architectures, justifying their selection as the primary XAI lens for this research.

Explainability in AI-generated text detection (AGTD) is therefore crucial not only for building end-user trust but also for ensuring fairness and accountability, particularly when systems are applied to non-native English speakers or highly specialized content ***(Masih et al., 2025)***. This proposal advocates for a systematic review of the current state of explainable detection, moving beyond binary classification accuracy to examine the qualitative insights that XAI techniques provide. The research will explore the synergy between traditional stylometric analysis and modern transformer-based detectors, as exemplified by the **HULLMI** framework ***(Joshi et al., 2024)***, which serves as a key structural reference for this work.

2. BACKGROUND/LITERATURE REVIEW

A preliminary review of the literature reveals three dominant thematic clusters in explainable AI-generated text detection: stylometry and classical machine learning approaches, transformer-based deep learning methods, and multi-label attribution frameworks. Organizing findings across these themes exposes both consistent patterns and notable contradictions that motivate the proposed review.

Stylometry and Classical Machine Learning Early approaches relied on handcrafted linguistic features to distinguish human from AI-generated text. ***(Aditya Shah, 2023)***

demonstrated that stylistic indices — including syllable count, punctuation ratios, and Simpson's Index — carry significant discriminative power, with LIME and SHAP revealing which features most influence classification decisions. Similarly***, (McGovern et al., 2024)*** identified consistent LLM "fingerprints" through POS tag distributions and high-frequency token patterns, though their work relied more on informal visualization than rigorous XAI. ***(Przystalski et al., 2025)*** extended stylometric analysis to short-sample scenarios, with SHAP pinpointing grammatical standardization and atypical word frequencies as the most persistent indicators of AI authorship. Collectively, these studies affirm the enduring relevance of stylometry, particularly in resource-constrained or short-text settings, though they differ on which specific features are most discriminative — a tension that warrants systematic synthesis.

Transformer-Based Deep Learning The advent of transformer architectures shifted detection toward high-accuracy, embedding-based approaches. The HULLMI framework ***(Joshi et al., 2024)*** established that RoBERTa-Sentinel outperforms traditional classifiers while LIME provides token-level transparency, positioning explainability as a complement to rather than a trade-off with accuracy. ***(Mitrovi'c et al., 2023)*** reached a similar conclusion using DistilBERT with SHAP, uncovering word-level reasoning behind the classification of short ChatGPT-generated reviews. ***(Masih et al., 2025)*** further demonstrated that combining transformer embeddings with LIME and SHAP produces interpretable outputs without significant accuracy loss. However, a recurring limitation across these studies is their susceptibility to adversarial manipulation — ***(Mohammadi et al., 2025)*** specifically showed that XAI-informed token replacement can systematically evade transformer-based detectors, exposing a critical vulnerability that the field has yet to adequately address.

Multi-Label Attribution and Cross-Domain Generalization The most recent strand of research moves beyond binary classification toward source attribution and domain-specific detection. ***(Najjar et al., 2025)*** demonstrated that XAI feature importance profiles can create distinct "author signatures" for different LLMs including Llama, Gemini, and Claude, suggesting that explainability is not merely a transparency add-on but a core analytical tool for attribution. ***(Zahraa J. Mohammed Ali 2025)*** extended this to multi-label classification using transformer embeddings with LIME and SHAP, achieving granular differentiation across multiple AI sources. Domain-specific challenges are addressed by ***(Baharifar et al., 2025)*** in the medical misinformation context, while ***(Iqbal & Kashem, 2025)*** proposed a generalized ML framework for source identification across varied text types. ***(Adilazuarda et al., 2023)*** provides the broadest comparative perspective, noting that no single detection paradigm consistently dominates across domains — a finding that reinforces the need for the synthesizing review proposed here.

Literature Review - Comparative Table

Author(s) & Year	Title of the Study	Method Used	Dataset	Explainability Method	Key Findings / Contributions	Limitations
Aditya Shah et al. (2023)	Detecting and Unmasking AI-Generated Texts through Explainable AI using Stylistic Features	Stylometry + Traditional ML (SVM, Logistic Regression)	Custom dataset of human and AI-generated texts	LIME, SHAP	Used stylistic features (syllable count, punctuation ratios, Simpson's Index) to classify text. LIME and SHAP revealed which stylometric indices most drive classification decisions.	Limited dataset diversity; focused mainly on English text.
Mitrovic et al. (2023)	ChatGPT or Human? Detect and Explain. Explaining Decisions of ML Models for Detecting Short ChatGPT-generated Text	Deep Learning (DistilBERT)	Short ChatGPT-generated text reviews vs. human reviews	SHAP	SHAP integrated with DistilBERT to assign feature importance to specific tokens, uncovering the reasoning behind classification of short ChatGPT-generated reviews.	Limited to short texts; may not generalize to longer documents.
Adilazuarda et al. (2023)	Beyond Turing: A Comparative Analysis of Approaches for Detecting Machine-Generated Text	Comparative study (ML, Transformers, Stylometry)	Multiple benchmark corpora including multilingual datasets	Not explicitly specified	Comprehensive comparison of detection paradigms; highlights trade-offs between accuracy and interpretability. Noted that no single method dominates across all domains.	Limited focus on explainability; multilingual coverage is partial.
McGovern et al. (2024)	Your Large Language Models Are Leaving Fingerprints	Stylometry (POS tag distributions, frequent tokens)	Texts generated by multiple LLMs (GPT, Llama, etc.)	Attention visualization (informal)	LLMs leave consistent stylistic fingerprints through POS distributions and token frequencies. Focused more on visualization than formal XAI; results suggest model-specific linguistic patterns.	No formal XAI (LIME/SHAP); primarily exploratory visualization.
Joshi et al. (2024)	HULLMI: Human vs LLM Identification with Explainability	Transformer-based (RoBERTa-Sentinel) + Traditional ML (Random Forest, Naive Bayes)	HULLMI benchmark dataset (multi-source, multi-LLM)	LIME	Foundational framework combining traditional and deep learning detectors with LIME for transparency. RoBERTa-Sentinel achieved superior performance while LIME provided interpretable explanations for model decisions.	Primarily English-language; limited adversarial robustness testing.
Masih et al. (2025)	Classifying Human vs. AI Text with Machine	Transformer-based + Traditional ML	Mixed human and AI-generated text corpus	LIME, SHAP	Demonstrated that explainability techniques can visualize feature importance and token-level contributions. Showed how LIME and	Dataset scope may limit cross-domain generalizability.

Author(s) & Year	Title of the Study	Method Used	Dataset	Explainability Method	Key Findings / Contributions	Limitations
	Learning and Explainable Transformer Models				SHAP demystify opaque model decisions in AI text classification.	
Baharifar et al. (2025)	Interpretable AI for Classifying Human- and LLM-Generated Medical Misinformation with Multi-Modal Features	Multi-modal ML (text + other features)	Medical misinformation dataset	Interpretable AI / Feature attribution	Applied interpretable AI for domain-specific (medical) detection of LLM-generated misinformation. Multi-modal features improved accuracy while explainability increased stakeholder trust.	Domain-specific; limited generalizability to non-medical content.
Iqbal & Kashem (2025)	A Machine Learning Framework for Identifying Sources of AI-Generated Text	Ensemble ML	Multi-source AI-generated and human-written text	Feature importance (ML-based)	Proposed a framework capable of source attribution (identifying which LLM generated text) beyond binary classification. Demonstrated practical applicability across multiple LLM sources.	Scalability to newer LLMs not fully explored.
Mohammadi et al. (2025)	Explainability-Based Token Replacement on LLM-Generated Text	Transformer-based with adversarial analysis	LLM-generated text datasets	LIME, SHAP (token-level)	Novel approach using XAI explanations to guide token replacement, demonstrating vulnerability of detectors to explainability-informed adversarial attacks.	Highlights a security gap; practical defence mechanisms not yet proposed.
Najjar et al. (2025)	Leveraging Explainable AI for LLM Text Attribution: Differentiating Human-Written and Multiple LLM-Generated Text	Multi-label classification (Transformers + Ensemble)	Texts from multiple LLMs (Llama, Bard/Gemini, Claude, GPT) and humans	XAI-based feature importance profiles	XAI used to create distinct 'author profiles' for different LLMs. Feature importance patterns can differentiate between multiple AI sources, moving beyond binary to multi-label attribution.	May require frequent retraining as LLMs evolve.
Przystalski et al. (2025)	Stylometry Recognizes Human and LLM-Generated Texts in Short Samples	Stylometry	Short-text samples from humans and LLMs	SHAP (feature attribution)	Stylometry remains effective even in short-sample scenarios. SHAP revealed specific linguistic markers (grammatical standardization, word frequencies) that LLMs over-utilize as discriminative features.	Short-sample focus; less effective on longer, more varied text.
Zahraa J. Mohammed Ali et al. (2025)	Interpretable Multi-Label Classification of Human- and LLM-Generated Texts Using Transformer Embeddings and XAI	Transformer embeddings (multi-label classifier)	Multi-label human and LLM-generated text dataset	LIME, SHAP	Extended detection to multi-label classification covering multiple LLMs. Transformer embeddings combined with LIME/SHAP provided granular interpretability, showing which textual features distinguish each LLM.	Computationally intensive; scalability may be a challenge.

3. PROBLEM STATEMENT

The widespread deployment of LLMs has intensified the need for reliable detection of AI-generated text across high-stakes domains. Current methodologies, however, suffer from two compounding limitations that undermine their practical utility.

First, while transformer-based detectors such as RoBERTa and DistilBERT achieve state-of-the-art classification accuracy, their internal decision-making processes remain largely opaque. This "black box" nature makes it difficult for end-users — educators, editors, or legal professionals — to trust, audit, or contest their outputs ***(Mitrovi'c et al., 2023)***. Second, there is no consensus on which linguistic features — such as perplexity, burstiness, syntactic standardization, or stylistic markers — most reliably indicate AI authorship across different LLMs and domains ***(McGovern et al., 2024; Przystalski et al., 2025)***. This inconsistency introduces risk of systematic bias, particularly against non-native English speakers or authors of highly technical content, where AI-like fluency may be a natural writing characteristic rather than evidence of machine generation.

Compounding these issues, existing detection tools are evaluated almost exclusively on binary classification accuracy, with little attention to interpretability, robustness, or cross-domain transferability. No comprehensive review currently synthesizes explainable detection methods across multiple LLMs, text domains, and sample lengths — a gap that leaves researchers, educators, and policymakers without a unified, evidence-based reference for deploying transparent detection systems. This research directly addresses that gap by systematically reviewing how XAI techniques can be integrated into the detection pipeline to deliver both high accuracy and human-understandable justification.

4. OBJECTIVES

The primary objective of this proposed research is to provide a comprehensive evaluation of the current landscape of explainable human vs. LLM-generated text detection. Specific objectives include:

1. To **categorize** existing detection methods into traditional machine learning, deep learning, and stylometric paradigms, mapping their respective strengths and trade-offs in accuracy and interpretability ***(Adilazuarda et al., 2023; Joshi et al., 2024)***.

2. To **evaluate** the application of post-hoc explainability techniques — specifically LIME and SHAP — within text detection pipelines, assessing their effectiveness in identifying discriminative linguistic features ***(Aditya Shah, 2023; Zahraa J. Mohammed Ali 2025)***.

3. To **synthesize** reported findings on how detection models generalize across various LLMs (e.g., GPT-3.5, GPT-4, Llama, Gemini) and text domains (e.g., academic writing, medical misinformation, social media) ***(Baharifar et al., 2025; Iqbal & Kashem, 2025)***.

4. To **identify** limitations and underexplored areas in current explainable detection methods, including susceptibility to adversarial attacks, poor robustness in short-sample scenarios, and absence of standardized evaluation benchmarks ***(Mohammadi et al., 2025; Przystalski et al., 2025)***.

5. Research Questions

To guide the research, the following research questions have been formulated in direct alignment with the stated objectives:

- **RQ1:** How do traditional machine learning, deep learning, and stylometric approaches compare in terms of detection accuracy and interpretability, and what are the key trade-offs among these paradigms?
- **RQ2:** To what extent do post-hoc explainability techniques — specifically **LIME** and **SHAP** — effectively identify and communicate the discriminative linguistic features that distinguish LLM-generated text from human-written text?
- **RQ3:** What does the existing literature reveal about the generalizability of explainable detection models across different LLMs and text domains, and which contextual factors most influence detection performance?
- **RQ4:** What are the most consistently reported limitations of current explainable detection frameworks, and which areas — such as adversarial robustness, short-sample detection, and evaluation standardization — remain underexplored in the literature?

6. Scope

The proposed research will focus on scholarly articles published between **2023** and **2026**, capturing the most recent advancements in LLM technology and explainable detection strategies. This timeframe was selected to reflect the rapid evolution of both generative AI capabilities and corresponding detection research following the widespread adoption of ChatGPT and similar systems.

The scope includes:

- **Models:** LLMs such as the GPT series, Llama, Claude, Google Gemini, and specialized models used in research or medical text generation.
- **Techniques:** Fine-tuned transformers, ensemble models, stylistic feature extraction, and linguistic fingerprinting methods.
- **Explainability:** Local and global interpretability methods (LIME, SHAP), attention mechanism visualization, and feature attribution approaches.
- **Domains:** Academic writing, online reviews, medical information, and general-purpose corpora such as HC3 and M4.

The research will exclude non-textual AI generation tasks such as image or video detection, as these involve fundamentally different generative mechanisms and evaluation frameworks. The primary focus will be on English-language texts, as the majority of established benchmark datasets — including HC3, M4, and HULLMI — are English-centric, which reflects the current state of the literature rather than a deliberate linguistic exclusion. Multilingual studies will be included where they offer unique or transferable explainability insights ***(Adilazuarda et al., 2023)***.

7. METHODOLOGY FOR THE REVIEW

The research will follow a **Systematic Literature Review (SLR)** methodology guided by the **PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)** framework, ensuring transparency, reproducibility, and methodological rigor throughout the research process.

1. **Search Strategy:** Electronic databases including Google Scholar, SciSpace, ArXiv, and IEEE Xplore will be searched using keywords such as *"LLM text detection," "Explainable AI," "LIME/SHAP for text classification,"* and *"human vs. AI text identification."* Boolean operators (AND, OR) will be applied to refine search results and maximize relevant coverage.
2. **Selection Criteria:** Papers will be screened in two stages — title/abstract screening followed by full-text review. Inclusion criteria require that papers: (a) are published between 2023 and 2026, (b) address AI-generated text detection, and (c) incorporate or discuss explainability methods. Papers will be excluded if they focus solely on non-textual modalities or lack a clearly described methodology.
3. **Data Extraction:** For each selected paper, data will be extracted regarding the detection architecture, explainability tool used, datasets employed, evaluation metrics reported, and primary linguistic findings. A standardized extraction template will be used to ensure consistency across all reviewed studies.
4. **Synthesis and Analysis**: A thematic comparative analysis will be conducted to identify trends — such as the shift from hand-crafted stylometric features to transformer embeddings — and to surface commonalities or contradictions in explainability findings across studies.
5. **Quality Assessment:** Each source will be evaluated against the following measurable criteria: (a) published in a peer-reviewed journal or recognized conference venue, (b) dataset size of no fewer than 500 samples, (c) inclusion of at least one standard evaluation metric (accuracy, F1-score, AUC), and (d) availability of replication materials such as code or data where applicable.

8. SIGNIFICANCE

The significance of this research extends across multiple stakeholder communities, each facing distinct but interconnected challenges posed by the proliferation of AI-generated text.

For the **academic research community**, this research provides a centralized, up-to-date synthesis of the rapidly evolving AGTD field, documenting the transition from opaque "black box" classifiers toward transparent, interpretable "glass box" detection systems. By consolidating findings across detection paradigms and explainability frameworks, it offers a structured reference that reduces redundancy in future research efforts.

For **educators and academic institutions**, the emphasis on explainability offers a pathway toward more justifiable and contestable plagiarism detection tools. A detection system that can articulate why a submission is flagged — citing specific linguistic patterns — is far more defensible in academic integrity proceedings than one that delivers a binary verdict without justification.

For **system developers and NLP researchers**, the identification of specific linguistic markers that LLMs consistently over-utilize — such as grammatical standardization or atypical word frequency distributions revealed through SHAP ***(Przystalski et al., 2025)*** — provides actionable insights for building more robust and adversarially resilient detection systems.

More broadly, as AI-generated content increasingly intersects **with public information, journalism, and policy**, the ability to detect and explain AI authorship in a transparent and auditable manner becomes a matter of societal trust. This research contributes a scientifically grounded foundation toward that goal, supporting the responsible deployment of detection tools in high-stakes environments.

9. CONCLUSION

The detection of LLM-generated text represents one of the most pressing challenges in contemporary Natural Language Processing, yet its value in high-stakes domains is fundamentally contingent on transparency. A detection system that cannot explain its decisions is of limited utility to educators, publishers, or policymakers who must justify and defend their reliance on such tools. This proposal has outlined a structured and rigorous plan for a Systematic Literature Review that addresses this critical need by examining the intersection of AI-generated text detection and eXplainable Artificial Intelligence.

By following the PRISMA framework and applying consistent quality assessment criteria, the proposed research will move beyond isolated findings to produce a thematically organized synthesis of the current state of explainable detection. It will document how the field has evolved from stylometric approaches to transformer-based architectures, evaluate the practical utility of LIME and SHAP across diverse detection contexts, and identify the most significant limitations and open research questions that remain unresolved.

The ultimate contribution of this research is twofold. First, it will serve as a consolidated reference for researchers, developers, and practitioners navigating a fragmented and rapidly evolving literature. Second, by surfacing consistent linguistic indicators of AI authorship and exposing gaps in adversarial robustness and cross-domain generalizability, it will provide actionable directions for the next generation of transparent, trustworthy, and practically deployable detection systems — contributing meaningfully to the broader goal of responsible AI governance in the age of generative models.

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